(a) VQE ansatz

```
4  n = 3
5  layers = 2
6  rng = np.random.default_rng(7)
7
8  qc = QuantumCircuit(n + 1, n + 1)
9
10 for layer in range(layers):
11     for qubit in range(n):
12         qc.ry(float(rng.uniform(0, np.pi)), qubit)
13         qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
14     for qubit in range(n - 1):
15         qc.cx(qubit, qubit + 1)
16
17 qc.h(n)
18 qc.measure(n, n)
19 with qc.if_test((qc.clbits[n], 1)):
20     qc.x(0)
```

data, classical or quantum

classical control, loop or measured bit

quantum via a control, not in classical PDG

classical PDG and def-use only, skips a CNOT

kept by every rule, def-use included

added by the classical PDG

added by the QDG

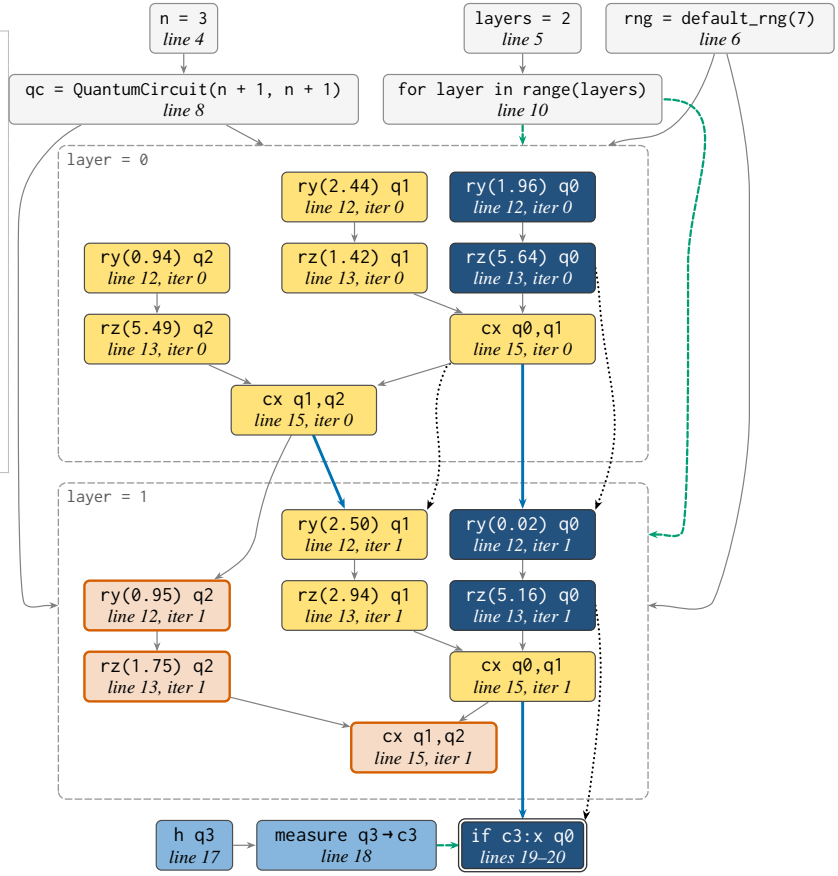
added by the QPDG, not a dependence

last write to the criterion

classical statement, all in the QDG slice

one iteration of the outer loop, with the inner loops of lines 11 and 14 unrolled

(b) Dependence graph of the four rules, criterion q_0



slices of q_0, of 19 operations:
def-use 5, $F = 0.55$ · classical PDG 7, $F = 0.97$
QDG 16, exactly the dependences, $F = 1.00$ · QPDG 19, $F = 1.00$